UNIVERSITÉ DE GENÈVE

IMAGERIE NUMÉRIQUE
13X004

TP 9: FT of typical signals

Author: Ethan Icet

E-mail: ethan.icet@etu.unige.ch

April 2021

UNIVERSITÉ DE GENÈVE
FACULTÉ DES SCIENCES
Département d'informatique

Exercise 1. Fourier Transform properties

In this exercise, we will prove formulas from page 102 Theme 7 of the course.

Name	Signal		Transform
impulse	$\delta(x)$	$\Leftrightarrow$	1
shifted impulse	$\delta(x-u)$	$\Leftrightarrow$	$e^{-j\omega u}$
box filter	$\mathrm{box}(x/a)$	$\Leftrightarrow$	$a\mathrm{sinc}(a\omega)$
tent	$\mathrm{tent}(x/a)$	$\Leftrightarrow$	$a\mathrm{sinc}^2(a\omega)$
Gaussian	$G(x;\sigma)$	$\Leftrightarrow$	$\frac{\sqrt{2\pi}}{\sigma}G(\omega;\sigma^{-1})$
Laplacian of Gaussian	$(\frac{x^2}{\sigma^4}-\frac{1}{\sigma^2})G(x;\sigma)$	$\Leftrightarrow$	$-\frac{\sqrt{2\pi}}{\sigma}\omega^2 G(\omega;\sigma^{-1})$
Gabor	$\cos(\omega_0 x)G(x;\sigma)$	$\Leftrightarrow$	$\frac{\sqrt{2\pi}}{\sigma}G(\omega\pm\omega_0;\sigma^{-1})$
unsharp mask	$(1+\gamma)\delta(x)$ $-\gamma G(x;\sigma)$	$\Leftrightarrow$	$(1+\gamma)-$ $\frac{\sqrt{2\pi}\gamma}{\sigma}G(\omega;\sigma^{-1})$
windowed sinc	$\mathrm{rcos}(x/(aW))$ $\mathrm{sinc}(x/a)$	$\Leftrightarrow$	(see Figure 3.29)

Recall Fourier Transform of Gaussian filter

The Gaussian filter is defined by:

$$g(t,\sigma)=\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{t^2}{2\sigma^2}}$$

$$\begin{aligned}
G(\omega)=\mathscr{F}\big(g(t,\sigma)\big)&=\int_{-\infty}^{\infty}g(t,\sigma)e^{-j\omega t}dt\\
\Rightarrow\frac{\mathrm{d}G(\omega)}{\mathrm{d}\omega}&=\frac{\mathrm{d}}{\mathrm{d}\omega}\left(\int_{-\infty}^{\infty}g(t,\sigma)e^{-j\omega t}dt\right)\\
&=\int_{-\infty}^{\infty}g(t,\sigma)(-jt)\cdot e^{-j\omega t}dt\\
&=\int_{-\infty}^{\infty}j\cdot\sigma^2\cdot\underbrace{\left(-\frac{t}{\sigma^2}\right)g(t,\sigma)}_{\frac{\mathrm{d}g(t,\sigma)}{\mathrm{d}t}}\cdot e^{-j\omega t}dt\\
&=j\sigma^2\left(\left[g(t,\sigma)e^{-j\omega t}\right]_{-\infty}^{\infty}-\int_{-\infty}^{\infty}g(t,\sigma)(-j\omega)\cdot e^{-j\omega t}dt\right)\\
&=j^2\sigma^2\omega\int_{-\infty}^{\infty}g(t,\sigma)\cdot e^{-j\omega t}dt\\
\frac{\mathrm{d}G(\omega)}{\mathrm{d}\omega}&=-\sigma^2\omega G(\omega)\\
\Rightarrow G(\omega)&=G(0)e^{-\omega^2\sigma^2/2}
\end{aligned}$$

$$\begin{cases}
G(0)=\displaystyle\int_{-\infty}^{\infty}g(t,\sigma)e^{-j\cdot 0\cdot t}dt=\int_{-\infty}^{\infty}g(t,\sigma)dt=1\\
e^{-\omega^2\sigma^2/2}=\frac{\sqrt{2\pi}}{\sigma}\cdot\frac{1}{\sqrt{2\pi(\sigma^{-1})^2}}e^{-\frac{\omega^2}{2(\sigma^{-1})^2}}=\frac{\sqrt{2\pi}}{\sigma}g(\omega,\sigma^{-1})
\end{cases}$$

Finally

$$\mathscr{F}\big(g(t,\sigma)\big)=\frac{\sqrt{2\pi}}{\sigma}g(\omega,\sigma^{-1})$$

a) $\left(\frac{t^2}{\sigma^4} - \frac{1}{\sigma^2}\right) \cdot g(t,\sigma) \overset{\mathscr{F}}{\Longrightarrow} -\frac{\sqrt{2\pi}}{\sigma}\omega^2 g(\omega,\sigma^{-1})$

We compute the Laplacian of Gaussian

$$\begin{aligned}
\frac{\mathrm{d}^2}{\mathrm{d}t^2} g(t,\sigma) &= \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}}{\mathrm{d}t} g(t,\sigma)\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{t^2}{2\sigma^2}}\right) \\
&= \frac{\mathrm{d}}{\mathrm{d}t}\left(-\frac{2t}{2\sigma^2}\cdot\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{t^2}{2\sigma^2}}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(-\frac{t}{\sigma^2}\cdot g(t,\sigma)\right) \\
&= \frac{\mathrm{d}}{\mathrm{d}t}\left(-\frac{t}{\sigma^2}\right)\cdot g(t,\sigma) - \frac{t}{\sigma^2}\cdot\frac{\mathrm{d}}{\mathrm{d}t}\big(g(t,\sigma)\big) \\
&= \left(\frac{t^2}{\sigma^4} - \frac{1}{\sigma^2}\right)\cdot g(t,\sigma)
\end{aligned}$$

We recall the time derivative property of the FT:

$$\mathscr{F}\left(\frac{\mathrm{d}^n}{\mathrm{d}t^n} f(t)\right) = (j\omega)^n F(\omega)$$

For $n = 2$ and $f(t) = g(t,\sigma)$ we find

$$\begin{aligned}
\mathscr{F}\left(\frac{\mathrm{d}^2}{\mathrm{d}t^2} g(t,\sigma)\right) &= (j\omega)^2 \mathscr{F}\big(g(t,\sigma)\big) \\
\mathscr{F}\left(\left(\frac{t^2}{\sigma^4} - \frac{1}{\sigma^2}\right) g(t,\sigma)\right) &= -\omega^2 \cdot \frac{\sqrt{2\pi}}{\sigma} g(\omega,\sigma^{-1})
\end{aligned}$$

b) $cos(\omega_0 t) g(t,\sigma) \overset{\mathscr{F}}{\Longrightarrow} \frac{\sqrt{2\pi}}{2\sigma}\big(g(\omega-\omega_0,\sigma^{-1}) + g(\omega+\omega_0,\sigma^{-1})\big)$

A Gabor filter is the product of a Gaussian signal with a cosine signal:

$$k(t) = cos(\omega_0 t)\cdot g(t,\sigma)$$

We compute the FT of a cosine signal:

$$\begin{aligned}
\mathscr{F}(cos(\omega_0 t)) &= \frac{1}{2}\mathscr{F}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right) = \frac{1}{2}\mathscr{F}\left(e^{j\omega_0 t}\right) + \frac{1}{2}\mathscr{F}\left(e^{-j\omega_0 t}\right) \\
&= \pi\delta(\omega-\omega_0) + \pi\delta(\omega+\omega_0)
\end{aligned}$$

We recall the product theorem of the FT:

$$\mathscr{F}\big(f(t)h(t)\big) = \frac{1}{2\pi} F(\omega) * H(\omega)$$

And the dirac delta function sifting/sampling property

$$f(t) * \delta(t-a) = \int_{-\infty}^{\infty} f(\tau)\delta(t-a-\tau)d\tau = f(t-a)$$

Finally

$$\begin{aligned}
\mathscr{F}\big(cos(\omega_0 t) g(t,\sigma)\big) &= \frac{1}{2\pi}\mathscr{F}(cos(\omega_0 t)) * \mathscr{F}\big(g(t,\sigma)\big) \\
&= \frac{1}{2\pi}\cdot\pi(\delta(\omega-\omega_0) + \delta(\omega+\omega_0)) * \frac{\sqrt{2\pi}}{\sigma} g(\omega,\sigma^{-1}) \\
&= \frac{\sqrt{2\pi}}{2\sigma}(g(\omega-\omega_0,\sigma^{-1}) + g(\omega+\omega_0,\sigma^{-1}))
\end{aligned}$$

c) $(1+\gamma)\delta(t) - \gamma g(t,\sigma) \overset{\mathscr{F}}{\Longrightarrow} (1+\gamma) - \gamma\frac{\sqrt{2\pi}}{\sigma} g(\omega,\sigma^{-1})$

An unsharp mask is given by the following formula:

$$u(t) = (1+\gamma)\delta(t) - \gamma g(t,\sigma)$$

We recall the linearity property of the FT:

$$\mathscr{F}\big(af_1(t) + bf_2(t)\big) = a\mathscr{F}\big(f_1(t)\big) + b\mathscr{F}\big(f_2(t)\big) = aF_1(\omega) + bF_2(\omega)$$

And the FT of the delta function:

$$\begin{aligned}\mathscr{F}(\delta(t)) &= \int_{-\infty}^{\infty} \delta(t)e^{-j\omega t}dt \\ &= 1\end{aligned}$$

We find

$$\begin{aligned}\mathscr{F}\big((1+\gamma)\delta(t) - \gamma g(t,\sigma)\big) &= (1+\gamma)\mathscr{F}(\delta(t)) - \gamma\mathscr{F}\big(g(t,\sigma)\big) \\ \mathscr{F}(u(t)) &= (1+\gamma) - \gamma\frac{\sqrt{2\pi}}{\sigma} g(\omega,\sigma^{-1})\end{aligned}$$

d) $\Lambda(t) \overset{\mathscr{F}}{\Longrightarrow} sinc^2(\omega/2)$

We consider the symmetric square pulse $\Pi(t)$ of length 1 and amplitude 1:

$$\Pi(t) = \begin{cases} 1 & \text{if } t \in [-1/2, 1/2] \\ 0 & \text{otherwise} \end{cases}$$

Also defined as (wikipedia):

$$rect(t) = \Pi(t) = \begin{cases} 1 & \text{if } |t| < 1/2 \\ 1/2 & \text{if } |t| = 1/2 \\ 0 & \text{if } |t| > 1/2 \end{cases}$$

We compute its FT:

$$\begin{aligned}\mathscr{F}(\Pi(t)) &= \int_{-\infty}^{\infty} \Pi(t)e^{-j\omega t}dt \\ &= \int_{-\infty}^{-1/2} 0\cdot e^{-j\omega t}dt + \int_{-1/2}^{1/2} 1\cdot e^{-j\omega t}dt + \int_{1/2}^{\infty} 0\cdot e^{-j\omega t}dt \\ &= \left[-\frac{1}{j\omega}e^{-j\omega t}\right]_{-1/2}^{1/2} \\ &= \frac{1}{j\omega}\left(e^{j\omega/2} - e^{-j\omega/2}\right) \\ &= \frac{2}{\omega} sin(\omega/2) \\ &= \frac{sin(\omega/2)}{\omega/2} \\ &= sinc(\omega/2)\end{aligned}$$

We define $\Lambda(t) := (\Pi * \Pi)(t)$

$$\begin{aligned}\Lambda(t) &= \int_{-\infty}^{\infty} \Pi(\tau)\Pi(t-\tau)d\tau \\ &= \int_{-\infty}^{-1/2} 0\cdot\Pi(t-\tau)d\tau + \int_{-1/2}^{1/2} \Pi(\tau)\Pi(t-\tau)d\tau + \int_{1/2}^{\infty} 0\cdot\Pi(t-\tau)d\tau\end{aligned}$$

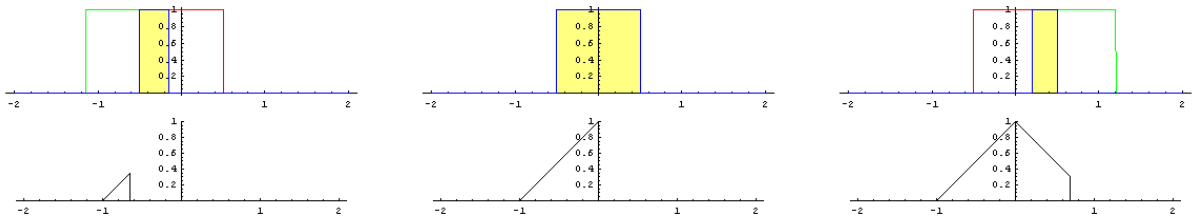
We have $\tau \in [-1/2, 1/2]$ we are interested in $\Pi(t-\tau) \neq 0 \Rightarrow (t-\tau) \in [-1/2, 1/2]$ and so $t \in [-1, 1]$

$$
\begin{aligned}
&= \begin{cases} \int_{t-1/2}^{1/2} \Pi(\tau)\Pi(t-\tau)d\tau & 0 < \mathrm{t} < 1 \\ \int_{-1/2}^{t+1/2} \Pi(\tau)\Pi(t-\tau)d\tau & \text{-1} < \mathrm{t} < 0 \\ 0 & \text{otherwise} \end{cases} \\
&= \begin{cases} \int_{t-1/2}^{1/2} 1\, d\tau & 0 < \mathrm{t} < 1 \\ \int_{-1/2}^{t+1/2} 1\, d\tau & \text{-1} < \mathrm{t} < 0 \\ 0 & \text{otherwise} \end{cases} \\
&= \begin{cases} \left[t\right]_{t-1/2}^{1/2} & 0 < \mathrm{t} < 1 \\ \left[t\right]_{-1/2}^{t+1/2} & \text{-1} < \mathrm{t} < 0 \\ 0 & \text{otherwise} \end{cases} \\
&= \begin{cases} 1/2 - t + 1/2 & 0 < \mathrm{t} < 1 \\ t + 1/2 - (-1/2) & \text{-1} < \mathrm{t} < 0 \\ 0 & \text{otherwise} \end{cases} \\
\Lambda(t) &= \begin{cases} 1 - |t| & \text{if } 0 < |t| < 1 \\ 0 & \text{otherwise} \end{cases}
\end{aligned}
$$

It corresponds to the triangular (/hat/tent) function seen in the TP8:

$$
\begin{aligned}
tri(x) = \Lambda(t) &= max(1 - |x|, 0) \\
&= \begin{cases} 1 - |t| & \text{if } |t| < 1 \\ 0 & \text{otherwise} \end{cases}
\end{aligned}
$$

Intuition: When we convolve the area increase at a linear rate until they're on top of each other then decrease similarly.



We recall the convolution theorem of the FT:

$$\mathscr{F}\big(f(t) * h(t)\big) = \mathscr{F}\big(f(t)\big) \cdot \mathscr{F}(h(t)) = F(t)H(t)$$

Finally

$$
\begin{aligned}
\mathscr{F}(\Lambda(t)) &= \mathscr{F}((\Pi * \Pi)(t)) \\
&= \mathscr{F}(\Pi(t))\mathscr{F}(\Pi(t)) = sinc(\omega/2)sinc(\omega/2) \\
\mathscr{F}(\Lambda(t)) &= sinc^2(\omega/2)
\end{aligned}
$$

Exercise 2. Continuous signal filtering

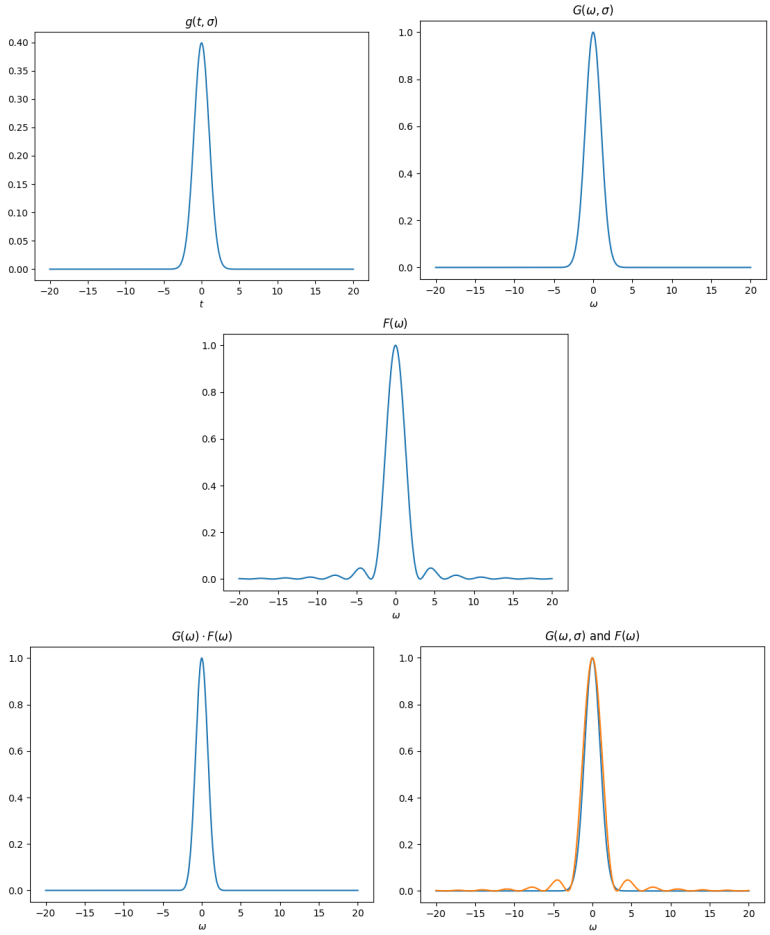
Filtering a signal $f(t)$ with a filter $h(t)$ consists in computing the convolution:

$$(f * h)(t) = \int_{-\infty}^{\infty} f(\tau) \cdot h(t-\tau) d\tau$$

Using the convolution theorem for FT, we can see that filtering corresponds to multiplication in the frequency domain: $F(f * h)(\omega) = F(\omega) \cdot H(\omega)$.

We will visualize different filtering on the signal $F(\omega) = sinc^2(\omega)$.

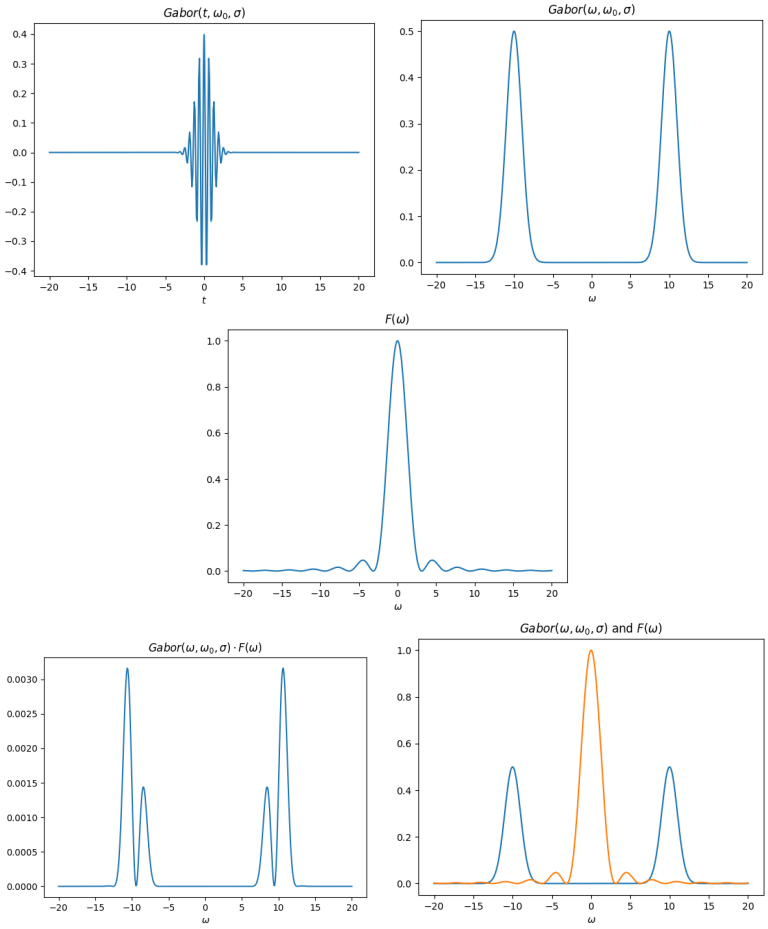
a)



Gaussian filtering ($\sigma = 1$)

In the frequency domain the lowest frequencies are close to 0 and the farther we get the higher they're. "Filtering a signal with a Gaussian blur kills high frequencies" because the non zero area is in the neighborhood of 0 which will be kept while the other frequencies are nullified.

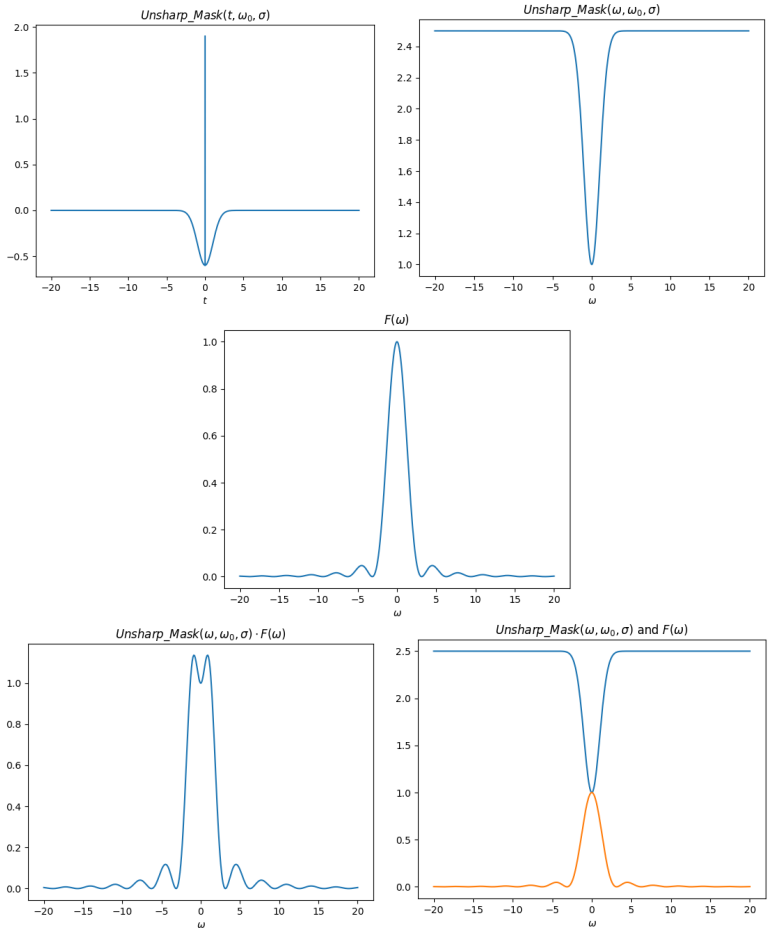
b)



Gabor filtering ($\omega_0 = 10$)

The Gabor filter corresponds to 2 Gaussians curve at $\pm\omega_0$ in the frequency domain. "Filtering with a Gabor filter at frequency ω_0 only keeps the parts of the signal which have a frequency close to ω_0." because the non zero areas are in the neighborhood of $\pm\omega_0$ which will be kept while the other frequencies are nullified.

c)



Unsharp mask filtering ($\gamma = 1.5$)

The Unsharp mask in the frequency domain is constant $(1+\gamma)$ except in the neighborhood of 0 where it decreases following an upside down Gaussian until 1. "The Unsharp mask filtering sharpens the signal by reducing low frequency components" because the signal is kept ($\geq 1\ \forall\omega$) but the high frequencies are amplified ($\times(1+\gamma)$) relatively to the low frequencies in the neighborhood of 0 ($\times \sim 1$)

Exercise 3. The Dirac comb (part 2)

The Dirac comb or train of impulses is the following "function":

$$Ш_T(t) = \sum_{k\in\mathbb{Z}\sigma(t-k\cdot T)}$$

Remember from TP7, that sampling a function at time interval T is the same as multiplying it with $Ш_T(t)$.

a)

Given any function $f(t)$ defined on $\mathbb{R}$, show that

$$(f * Ш_T)(t) = \sum_{k\in\mathbb{Z}} f(t-kT)$$

Explain why this function is periodic of period T.
Reminder: The delta function has the property that $f(t) * \delta(t-y) = f(t-y)$.

b)

Illustrate point a) with the hat function:

$$h(t) = \begin{cases} 1-|t| & \text{if } |t| \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

For various $T = 1,\ 1.5,\ 2,\ 4$, plot on the same graph:

- The shifted functions $h(t-kT)$ on an interval $t \in [-5, 5]$.
- The periodic signal $(h * Ш_T)(t)$

c)

You've seen in the course that

- $\mathscr{F}(Ш_T) = Ш_{\frac{1}{T}}$
- $\mathscr{F}(f * g) = \mathscr{F}(f) \cdot \mathscr{F}(g)$
- $\mathscr{F}(f \cdot g) = \frac{1}{2\pi}\mathscr{F}(f) * \mathscr{F}(g)$

Use these facts and point a) to explain the following:

i. The Fourier transform of a periodic function is discrete (non-zero only at isolated points). Give an example.

ii. The Fourier transform of a discrete signal is a periodic function. Give an example.

iii. If $f(t)$ is both discrete and periodic, then its transform is also discrete and periodic and vice-versa. Give an example.

Remark: This last result plays a central role in the definition of the DFT.